

Biochemistry Lab #6
Blood Glucose and Cholesterol Enzymatic Assays
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Introduction:

Cholesterol and blood glucose levels are used as diagnostic tools to indicate a variety of chronic diseases that are prevalent causes of mortality worldwide. Enzymatic assays are commonly used to identify the relative ranges of these biomolecules because they are highly specific, sensitive, and easy to perform. These assays rely on enzyme-catalyzed reactions that produce a colored product whose intensity is proportional to the analyte concentration. By comparing the color intensity of an unknown sample to that of a known standard, the concentration of both glucose and cholesterol in blood can be accurately measured.

The blood glucose (Patient C) and cholesterol (Patient E) concentrations were determined for an unknown patient using enzymatic assays. These assays were modified from clinical procedures for manual spectrophotometric analysis. The reagents were sourced from Thermo Electron (Louisville, CO), with the glucose being measured at 340 nm, and the cholesterol at 500 nm using a Shimadzu MiniUV. Blood glucose levels are an important indicator of metabolic health, with elevated concentrations strongly associated with diabetes. Cholesterol levels are used to assess cardiovascular health. Hypercholesterolemia is a risk factor for heart disease, stroke, and atherosclerosis.

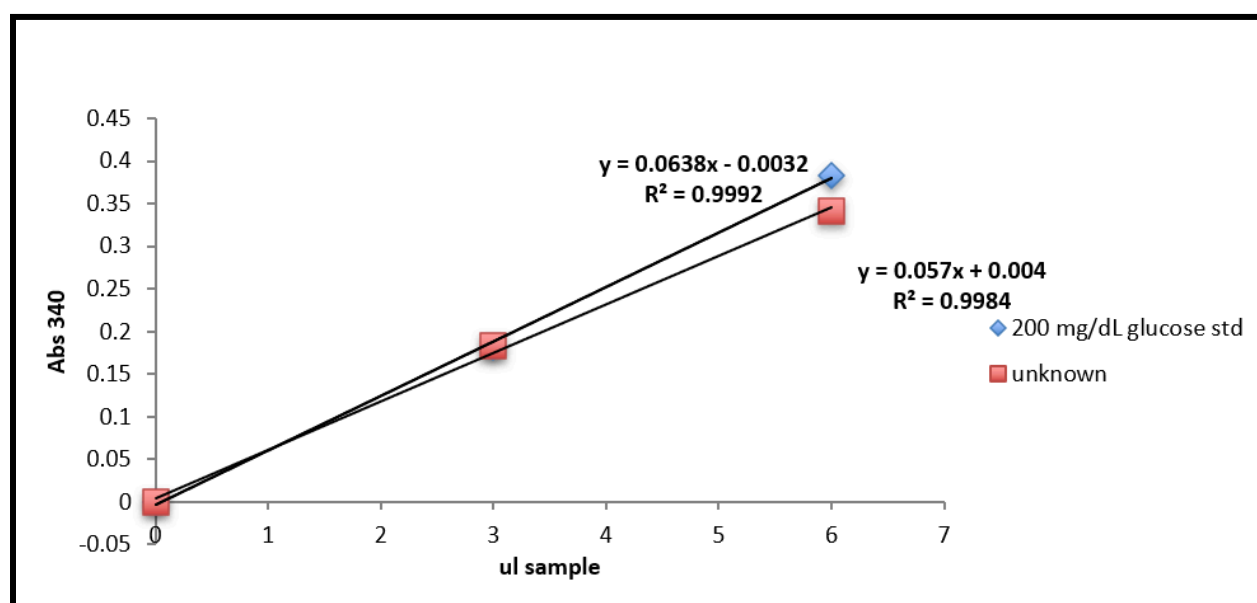
Results:

The blood glucose concentration for patient C was determined. A glucose standard concentration of 200 mg/dL was used. The final concentration was computed from Table 1, with a resulting glucose concentration of 179 mg/dL or 9.94 mmol/L.

Table 1: Glucose Assay for Patient C

Sample	Volume of Sample (μL)	Reagent Volume (mL)	Absorbance (340 nm)
Blank	—	0.90	0.000
Standard 1	3	0.90	0.182
Standard 2	6	0.90	0.383
Unknown 1	3	0.90	0.183
Unknown 2	6	0.90	0.342

Graph 1: Graph of Glucose Assay (Abs 340 vs μL Sample) for Patient C.



The cholesterol concentration for patient E was determined. A cholesterol standard concentration of 300 mg/dL was utilized. The final concentration was computed from Table 2, with a resulting cholesterol concentration of 228 mg/dL or 5.90 mmol/L.

Table 2: Cholesterol Assay for Patient E

Sample	Volume of Sample (μL)	Reagent Volume (mL)	Absorbance (500 nm)
Blank	—	1.00	0.000
Standard 1	5	1.00	0.218
Standard 2	10	1.00	0.392
Unknown 1	5	1.00	0.164
Unknown 2	10	1.00	0.298

Graph 2: Graph of Cholesterol Assay (Abs 500 vs μL Sample) for Patient E

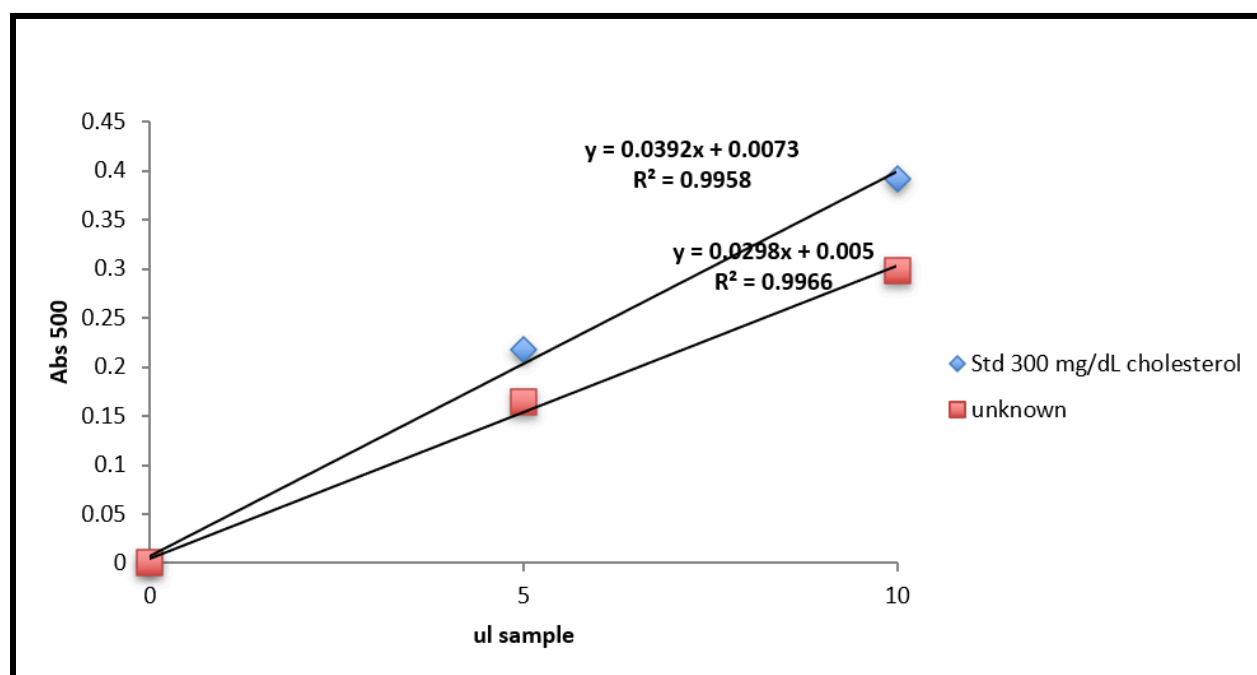


Table 3: Summary of Glucose and Cholesterol Assays

Analyte	Patient ID	Concentration (mg/dL)	Concentration (mmol/L)	Reference Range (mg/dL)	Interpretation
Glucose	C	179	9.94	70-100	Abnormal
Cholesterol	E	228	5.90	<200	Abnormal

Discussion:

Both enzymatic assays successfully resulted in a measurable response proportional to the concentration. Both the glucose and cholesterol assays presented a linear line of best fit in line with the sample volume, which is the expected conclusion given Beer's Law.

In the glucose assay, the absorbance was taken at 340 nm and scaled linearly, indicating that the enzymes were reliably oxidizing glucose. The calculated concentration was 179 mg/dL or 9.94 mmol/L which is significantly higher (by a factor of ~1.8) than the reference range provided by the Mayo Clinic of 70-100 mg/dL. This high glucose level is indicative of **patient C** likely being hyperglycemic, diabetic, or incorrectly fasted before the blood test.

The cholesterol assay also exhibited a linear scaling at 500 nm. The determined cholesterol concentration was 228 mg/dL or 5.90 mmol/L. According to the Mayo Clinic, the typical total blood cholesterol level is below 200 mg/dL. Since the calculated range is higher than 200, this indicates that **patient E** has hypercholesterolemia, which is a risk factor for atherosclerosis, cardiovascular disease, and stroke.

Both enzymatic assays demonstrate utility in the ability to quickly determine potential diagnoses and quantify biomolecules.

Materials and Methods:

Materials and reagents for the glucose and cholesterol assay were obtained from Thermo Electron (Louisville, CO). The spectrophotometer used was a Shimadzu MiniUV.

For the glucose assay, each sample was prepared in a microfuge tube. A standard glucose solution (3 μL and 6 μL) and patient C's sample (3 μL and 6 μL) were mixed with 0.90 mL of the glucose reagent. These were incubated for 3 minutes at 37°C. Their absorbances were then measured at 340 nm compared to a blank.

The standard cholesterol solutions (5 μL and 10 μL) and patient E's sample (5 μL and 10 μL) were mixed with 1.00 mL of cholesterol reagent in microfuge tubes. These were incubated for 5 minutes at 37°C. Their absorbances were then measured at 500 nm compared to a blank.

The concentrations were determined through the construction of a calibration curve, then using linear regression to calculate the patient's concentration.

References:

Mayo Foundation for Medical Education and Research. (2024, March 27). *Diabetes*. Mayo Clinic.

<https://www.mayoclinic.org/diseases-conditions/diabetes/diagnosis-treatment/drc-20371451>

Mayo Foundation for Medical Education and Research. (2025, March 7). *High cholesterol*. Mayo Clinic.

<https://www.mayoclinic.org/diseases-conditions/high-blood-cholesterol/diagnosis-treatment/drc-20350806>